

Linear Algebra I: mechanics

Week 3 — Algorithmic Approaches to Mathematics

Dhruva Raman

Linear Algebra underpins everything

Linear Algebra underpins everything

What is Linear Algebra?

You'll know by the end of the course!

What do I need to learn?

Each time you learn this subject you get more out of it

Excessive content. Remember the cow.

Minimal goals

Conceptual **and mechanical** understanding
(inseparable)

- Dot products
- Matrix multiplication/algebra
- Correlation

Conceptual understanding

- Matrix equations
- Matrix inverses

Next lecture: geometrical understanding

This lecture needs friends

3blue1brown course

- amazing visualisations
- transcripts available!
- hard to pace/concentrate on computer 🙄

Notebooks

- Maths is not a spectator sport.

Buying groceries

3 apples. £1.50 each. Cost?

2 bananas. 75p each. Cost?

Total cost

$$[1.50 \ 0.75] \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 1.5 \times 3 + 0.75 \times 2 = 6$$

Variable quantities

$$[1.50 \ 0.75] \begin{bmatrix} \text{🍏} \\ \text{🍌} \end{bmatrix} \\ = 1.5 \text{🍏} + 0.75 \text{🍌}$$

Buying groceries

Abstracting to variable (n) quantities, items

$$\underbrace{[p_1 \ p_2 \ p_3 \ \dots \ p_n]}_{\text{prices}} \quad \underbrace{\begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ \vdots \\ g_n \end{bmatrix}}_{\text{quantities}}$$

$$= p_1 g_1 + p_2 g_2 + p_3 g_3 + \dots p_n g_n$$

- e.g. $n = 10$

$$\underline{p} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ \vdots \\ p_n \end{bmatrix}, \underline{g} = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ \vdots \\ g_n \end{bmatrix} \text{ are } \textbf{vectors}$$

- *Ordered* collections of numbers

Buying groceries

Abstracting to variable (n) quantities, items

$$\underbrace{[p_1 \ p_2 \ p_3 \ \dots \ p_n]}_{\text{prices}} \quad \underbrace{\begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ \vdots \\ g_n \end{bmatrix}}_{\text{quantities}}$$

$$= p_1 g_1 + p_2 g_2 + p_3 g_3 + \dots p_n g_n$$

- e.g. $n = 10$

Transposition (swap rows and columns)

$$\underline{p} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ \vdots \\ p_n \end{bmatrix} \text{ is a } \textit{column} \text{ vector}$$

$$\underline{p}^T = [p_1 \ p_2 \ p_3 \ \dots \ p_n] \text{ is a } \textit{row} \text{ vector}$$

Total cost is $\underline{p}^T \underline{g}$

“The dot product of $\underline{p}$ and $\underline{g}$ ”

Cycling with groceries

Total cost =

$$\begin{aligned}\underline{p}^T \underline{g} &= \underbrace{\begin{bmatrix} p_1 & p_2 & p_3 & \dots & p_n \end{bmatrix}}_{\text{prices}} \underbrace{\begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ \vdots \\ g_n \end{bmatrix}}_{\text{quantities}} \\ &= p_1 g_1 + p_2 g_2 + p_3 g_3 + \dots p_n g_n\end{aligned}$$

Total weight =

$$\begin{aligned}\underline{w}^T \underline{g} &= \underbrace{\begin{bmatrix} w_1 & w_2 & w_3 & \dots & w_n \end{bmatrix}}_{\text{weights}} \underbrace{\begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ \vdots \\ g_n \end{bmatrix}}_{\text{quantities}} \\ &= w_1 g_1 + w_2 g_2 + w_3 g_3 + \dots w_n g_n\end{aligned}$$

Vertically concatenate (vcat):

$$\overbrace{\begin{bmatrix} p_1 & p_2 & p_3 & \dots & p_n \\ w_1 & w_2 & w_3 & \dots & w_n \end{bmatrix}}^{\text{call this } M} \begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ \vdots \\ g_n \end{bmatrix} = \begin{bmatrix} \text{total cost} \\ \text{total weight} \end{bmatrix}$$

$$\underline{p}, \underline{g}, \underline{w} \in \mathbb{R}^n$$

$$M \in \mathbb{R}^{2 \times n}$$

$$M \underline{g} \in \mathbb{R}^2$$

Try for yourself

Cost/quantity of: $\begin{bmatrix} \text{🍏} \\ \text{🍌} \\ \text{📦} \end{bmatrix} ?$

$$\begin{bmatrix} \text{£1} & \text{£3} & \text{£2} \\ 4\text{lb} & 1\text{lb} & 3\text{lb} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} \text{£11} \\ 18\text{lb} \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 & 3 & 3 \\ 1 & 5 & 1 & 2 \\ 6 & 3 & 5 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 2 \\ 1 \end{bmatrix} = ?$$

Matrix Ordered collection of numbers *across two index dimensions* (rows and columns)

$$M \in \mathbb{R}^{m \times n} = \begin{bmatrix} M_{11} & M_{12} & \dots & M_{1n} \\ M_{21} & M_{22} & \dots & M_{2n} \\ \dots & \dots & \dots & \dots \\ M_{m1} & M_{m2} & \dots & M_{mn} \end{bmatrix}$$

where *elements* $M_{ij} \in \mathbb{R}$

Concatenated dot products are called *matrix multiplication*

Two sets of groceries

$\begin{bmatrix} \text{🍏} \\ \text{🍌} \end{bmatrix}$: Alice buys $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$, Bob buys $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$

$$MG = \begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 4.50 \\ 2 & 1.5 \end{bmatrix}$$

Horizontal concatenation:

(hcat): $G =_{\text{fruit}} \left\{ \overbrace{\begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}}^{\text{person}} \right.$

	Alice	Bob
cost?	£6	£4.50
weight?	2 lb	1.5 lb

Vertical concatenation:

prices $\rightarrow \begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix}$
weights $\rightarrow$

Rows of M ask questions

Answered for each column of G

We've done matrix multiplication

How to do matrix multiplication

1. Sketch out the shape

$$\begin{bmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{bmatrix} \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \\ \bullet & \bullet \end{bmatrix}$$

2. Divide left multiplier into rows.

Right into columns:

$$\begin{bmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{bmatrix} \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \\ \bullet & \bullet \end{bmatrix}$$

3. Result shape inherits *rows* of LHS,
cols of RHS

$$\begin{bmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{bmatrix} \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} = \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix}$$

1

¹Matmul **only possible** if LHS columns = RHS rows

How to do matrix multiplication

4. Fill in result. O_{ij} is dot product of row i (LHS) with col j (RHS)

$$\left[\begin{array}{ccc} \bullet & \bullet & \bullet \\ \hline \bullet & \bullet & \bullet \end{array} \right] \left[\begin{array}{c|c} \bullet & \bullet \\ \bullet & \bullet \\ \bullet & \bullet \end{array} \right] = \left[\begin{array}{cc} \bullet & \bullet \\ \bullet & \bullet \end{array} \right]$$

Practice

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 4 & 1 \\ 5 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

Practice

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} \neq \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

- Multiplication is **not** commutative:
order matters

Taking a step back

Functions transform numbers

Example 1

$$f(x) = x^2 \quad f : \mathbb{R} \rightarrow \mathbb{R}^+$$

Example 2

$$+(x, y) = x + y \quad f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$

Terminology

Domain: set of possible inputs

Range: set of possible outputs

$$f : \text{Domain} \rightarrow \text{Range}$$

Functions are transforms between more general (any) objects

Function 1: Image $\rightarrow$ Image



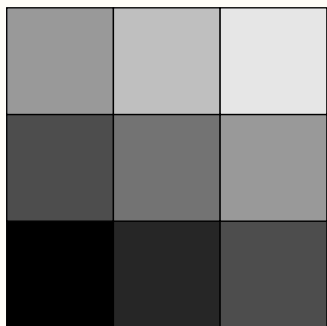
Function 2: Image $\rightarrow$ String



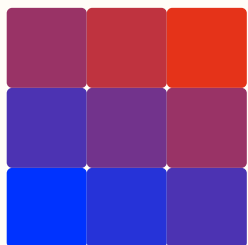
“seagull”

General things often reduce to arrays of numbers

Arrays are *ordered collections*...wait for more

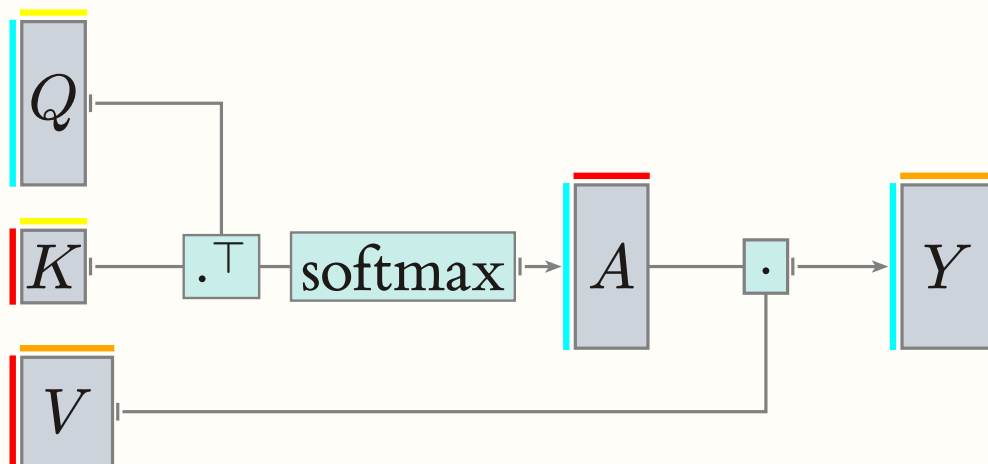


$$\begin{bmatrix} 0.6 & 0.75 & 0.9 \\ 0.3 & 0.45 & 0.6 \\ 0 & 0.15 & 0.3 \end{bmatrix}$$



$$\begin{bmatrix} (0.6, 0.2, 0.4) & (0.75, 0.2, 0.25) & (0.9, 0.2, 0.1) \\ (0.3, 0.2, 0.7) & (0.45, 0.2, 0.55) & (0.6, 0.2, 0.4) \\ (0.0, 0.2, 1.0) & (0.15, 0.2, 0.85) & (0.3, 0.2, 0.7) \end{bmatrix}$$

General things often reduce to arrays of numbers



$$\text{Attention}(Q, K, V) = \text{softmax}_{\text{row}}\left(\frac{QK^\top}{\sqrt{d}}\right)V$$

Single-head attention
(the basic unit of LLMs)

Every capitalised variable here is
a matrix

Query $Q \rightarrow$ Output Y

Running an LLM is just multiplying a load of matrices

Linear algebra?

Algebra $(+, -, \times, \dots)$ on arrays with weird shapes

Array shapes and dimensions

$$\underline{v} = \begin{bmatrix} 1 \\ 3 \\ 4 \\ 5 \end{bmatrix}$$

$$\text{np.shape}(v) = (4,)$$

$$\underline{v}_2 = 3$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$\text{np.shape}(A) = (2, 3)$$

$$A_{21} = 4$$

Tensor dimension of an array

is number of indices needed to locate elements

Vectors are arrays
of tensor dimension 1

Matrices are arrays of
tensor dimension 2

Confusing terminology **alert**

$$\underline{v} = [4 \ 2 \ 1]$$

Programmers

“ $\underline{v}$ is a 1d array”
(i.e. tensor dimension 1)

Mathematicians

“ $\underline{v}$ is a 3d vector”
(i.e. vector dimension 3)

1

¹Vector dimension definition next week

Confusing indexing **alert**

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

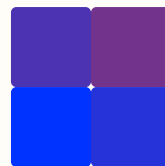
Julia programmers / mathematicians

“ $A_{12} = 2$ ”
(*counting*)

Python programmers

“ $A_{12} = 6$ ”
(*offset: start at 0*)

Tensor dimension?



$$\begin{bmatrix} (0.3, 0.2, 0.7) & (0.45, 0.2, 0.55) \\ (0.0, 0.2, 1.0) & (0.15, 0.2, 0.85) \end{bmatrix}$$

$$M \in \mathbb{R}^{2 \times 2 \times 3}$$

(Two rows, two columns, three *layers*)¹

¹Multilayer perceptrons are also 3d tensors

Netflix & ch...matrix multiplication

Ratings scale: $[-10, 10] \cap \mathbb{Z}$

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	-1	-1
Bob	1	1	-1	-1
Carol	-1	-1	1	1
Eve	1	-1	0	0
Frank	10	10	-10	-10

Alice likes a new film....

? Do you think Bob will like it

? Do you think Carol will like it

? Do you think Eve will like it

? Do you think Frank will like it

Alignments between watchers

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	-1	-1
Bob	1	1	-1	-1
Carol	-1	-1	1	1
Eve	1	-1	0	0
Frank	10	10	-10	-10

Dot product of Alice and Bob?

$$\underline{a}^T \underline{b} = [1 \ 1 \ -1 \ -1] \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} = 4$$

They are **aligned**. They **correlate**

- $\underline{a}^T \underline{c} < 0$: They are **anticorrelated**
- $\underline{a}^T \underline{e} = 0$: They are **uncorrelated**

? Can we get all the alignments quickly?

A grid of alignments

$$A = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & 0 & 0 \\ 10 & 10 & -10 & -10 \end{bmatrix}$$

rows are people,
columns are films

$$A^T?$$

columns are people,
rows are films

$$AA^T?$$

rows *and* columns are people

$$(AA^T)_{ij} ? \text{ e.g. } (AA^T)_{12}$$

Dot product between film preferences of
person i and person j

A grid of alignments

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	-1	-1
Bob	1	1	-1	-1
Carol	-1	-1	1	1
Eve	1	-1	0	0
Frank	10	10	-10	-10

$$AA^T = \begin{bmatrix} 4 & 4 & -4 & 0 & 40 \\ 4 & 4 & -4 & 0 & 40 \\ -4 & -4 & 4 & 0 & -40 \\ 0 & 0 & 0 & 2 & 0 \\ 40 & 40 & -40 & 0 & 400 \end{bmatrix}$$

Symmetric: $AA_{ij}^T = (AA^T)_{ji}$

Quickly summarises all different people alignments

A grid of alignments

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	-1	-1
Bob	1	1	-1	-1
Carol	-1	-1	1	1
Eve	1	-1	0	0
Frank	10	10	-10	-10

$$A^T A = \begin{bmatrix} 104 & 102 & -103 & -103 \\ 102 & 104 & -103 & -103 \\ -103 & -103 & 103 & 103 \\ -103 & -103 & 103 & 103 \end{bmatrix}$$

Symmetric: $A^T A_{ij} = (A^T A)_{ji}$

Summarises? All different *film* alignments

Is Frank a better fit for Alice than Bob?

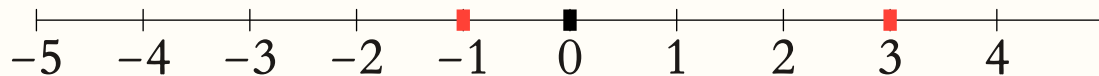
	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	-1	-1
Bob	1	1	-1	-1
Carol	-1	-1	1	1
Eve	1	-1	0	0
Frank	10	10	-10	-10

$$AA^T = \begin{bmatrix} 4 & 4 & -4 & 0 & 40 \\ 4 & 4 & -4 & 0 & 40 \\ -4 & -4 & 4 & 0 & -40 \\ 0 & 0 & 0 & 2 & 0 \\ 40 & 40 & -40 & 0 & 400 \end{bmatrix}$$

- No, stronger opinions, same alignment!
- $\underline{f}$ has larger *magnitude*, which amplifies the dot product

Next goal: measure of alignment that is magnitude-independent

The magnitude of a vector is called its norm



Numbers (magnitude)

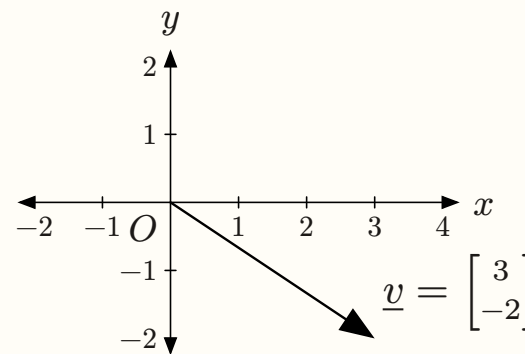
- $-3 < 1$ (ordering on number line)
- $|-3| > |1|$ (further from zero)

Definition: norm

- The norm of a vector is a notion of its *distance from the zero vector*
- Denoted $\left\| \begin{bmatrix} \cdot \\ \cdot \end{bmatrix} \right\|$, eg $\left\| \begin{bmatrix} -2 \\ 3 \end{bmatrix} \right\|$

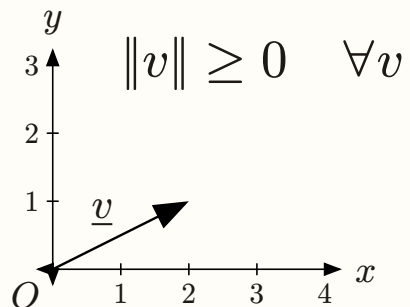
Vectors

- $\begin{bmatrix} 3 \\ -2 \end{bmatrix} < \begin{bmatrix} -2 \\ 2 \end{bmatrix}$? No ordering!



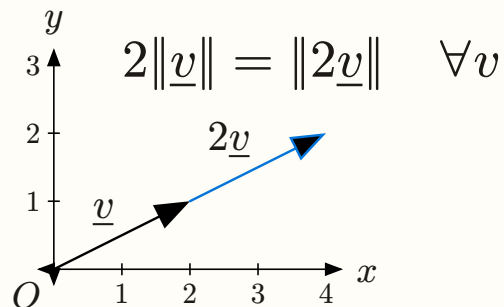
What properties should a norm have?

? Non-negativity



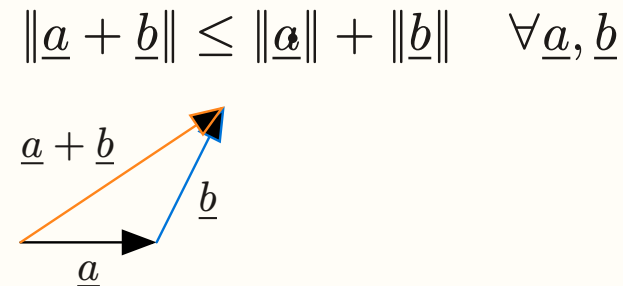
Distance from zero can't be negative!

? Scaling homogeneity



Twice the vector, twice the distance!

? Triangle inequality



Going *via* a waypoint is never faster!

Generalisation:

$$k\|\underline{v}\| = \|k\underline{v}\| \quad \forall k \in \mathbb{R}$$

There are lots of norms!

... but use these three 99% of the time...

L_1 (taxicab) norm

$$\|\underline{v}\|_1 = \sum_{i=1}^n |v_i|$$

L_2 (Euclidean) norm

$$\|\underline{v}\|_2 = \sqrt{\sum_{i=1}^n v_i^2}$$

L_∞ norm

$$\|\underline{v}\|_\infty = \max_i |v_i|$$

$$\underline{x} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$$

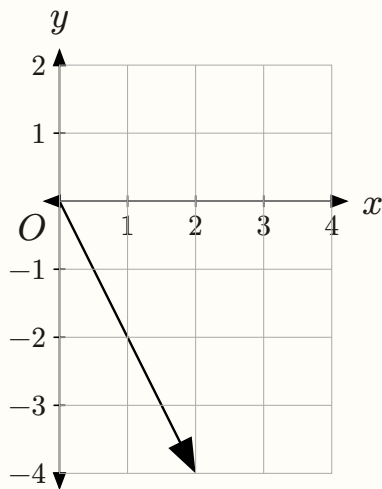
$$\|\underline{x}\|_1 = |2| + |-4| = 6$$

taxicab?

$$\|\underline{x}\|_2 = \sqrt{2^2 + (-4)^2} = \sqrt{20}$$

length of line?

$$\|\underline{x}\|_\infty = 4$$



Normalising vectors

Normalised vector notation

$$\hat{v} = \frac{v}{\|v\|}$$

? Which norm?

- Depends! (Usually L_2)

Practice

	Dune	Interstellar	Love actually	Mamma Mia
Alice	1	1	-1	-1
Bob	1	1	-1	-1
Carol	-1	-1	1	1
Eve	1	-1	0	0
Frank	10	10	-10	-10

Normalise Bob and Frank's rows (L_2)

? Which has higher *normalised* alignment? (correlation...)

¹Normalised vector often called a **unit** vector

Correlation is the dot product of unit vectors

$$\text{corr} \left(\begin{bmatrix} 2 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 4 \end{bmatrix} \right) ?$$

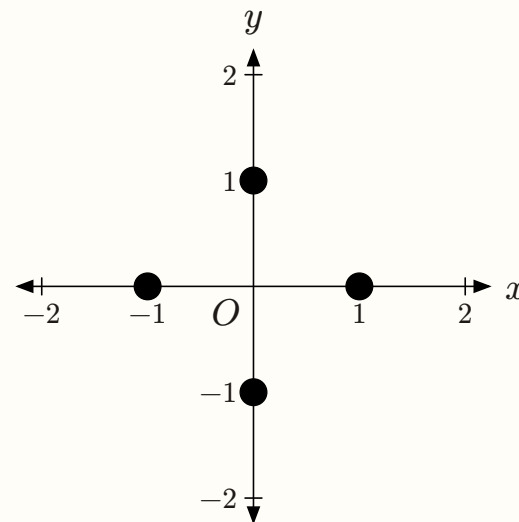
$$= \frac{-6-8}{\sqrt{8}\sqrt{25}} = \frac{-1.4}{\sqrt{2}}$$

? For next week

- Manually build a *correlation* matrix for the movie tastes
- Draw the **unit spheres** in L_1, L_2, L_∞

$$S = \{ \underline{v} \in \mathbb{R}^2 : \|\underline{v}\| = 1 \}$$

- Maximum/minimum possible correlation?



Hint: all marked points are on all spheres!

Completing the algebra

What is an algebra?

A framework for groups of objects to interact and **make babies**

Algebra on numbers

- Addition: $4 + 5 = 5 + 4 = 9$
- Multiplication: $4 \times 5 = 20$
- Associativity:
 $(4 \times 5) \times 6 = 4 \times (5 \times 6)$

Algebra on vectors/matrices

- Multiplication: ✓
- Addition: $\begin{bmatrix} 4 \\ 5 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 9 \end{bmatrix}$
(*elementwise*)
- Scaling: $3 \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 \\ 15 \end{bmatrix}$
- Associativity: $(AB)C = A(BC)$

Algebra cautions

Multiplying numbers is a commutative operation

$$a \times b = b \times a \quad \checkmark$$

Multiplying matrices is *not* a commutative operation

¹

$$AB \neq BA$$

? For next week

Construct two matrices in $\mathbb{R}^{2 \times 2}$ that do *not* commute under multiplication

¹ Addition is commutative in *all* algebras

Algebra cautions

You can multiply/add any
two numbers

You can only multiply/
add matrices with
compatible shapes

? For next week

If $A \in \mathbb{R}^{m \times n} \wedge B \in \mathbb{R}^{p \times q}$, specify conditions on $m, n, p, q \in \mathbb{N}$ that

- make AB an allowable operation
- make BA an allowable operation

Special matrices

The **additive identity**:

$$\underline{0}_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- adding this doesn't change things

$$A + \underline{0}_n = \underline{0}_n + A = A \\ \forall A \in \mathbb{R}^{n \times n}, n \in \mathbb{N}$$

The **multiplicative identity**:

$$\underline{I}_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- multiplying this doesn't change things

$$A \underline{I}_n = A \quad \text{and} \quad \underline{I}_m A = A \quad \forall A \in \mathbb{R}^{m \times n}$$

Law of Transposition

$$(AB)^T = B^T A^T$$

- You reverse the multiplication order when you transpose

? Demonstrate for yourself on

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 \\ 6 & 2 \\ 7 & 3 \end{bmatrix}$$

? Use the law to rewrite $(ABC)^T$

Solving matrix equations

Recap: Two sets of groceries

$\begin{bmatrix} \text{🍏} \\ \text{🍌} \end{bmatrix}$: Alice buys $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$, Bob buys $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$

$$MG = \begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 4.50 \\ 2 & 1.5 \end{bmatrix}$$

Horizontal concatenation:

(hcat): $G =_{\text{fruit}} \overbrace{\begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}}^{\text{person}}$

	Alice	Bob
cost?	£6	£4.50
weight?	2 lb	1.5 lb

Vertical concatenation:

prices $\rightarrow \begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix}$
weights $\rightarrow$

Rows of M ask questions

Answered for each column of G

Basket + questions $\xrightarrow{\text{matmul}}$ answers

Running matmul backwards

? + questions $\xrightarrow{\text{matmul}}$ answers

Questions

prices $\rightarrow \begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix}$
weights $\rightarrow$

Answers

	Alice	Bob
cost?	£6	£4.50
weight?	2 lb	1.5 lb

What was in baskets?

$$\begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 6 & 4.50 \\ 2 & 1.5 \end{bmatrix}$$

Running matmul backwards

This is called a matrix equation

$$\underbrace{A}_{\text{known}} \underbrace{X}_{\text{unknown}} = \underbrace{Y}_{\text{known}}$$

What was in baskets?

$$\begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 6 & 4.50 \\ 2 & 1.5 \end{bmatrix}^1$$

¹Note that X and Y could be matrices or vectors

How do we do this for numbers?

Forward

$$4 \times 3 = 12$$

Backward

$$4 \text{ 🍌 } = 12 \Rightarrow \text{ 🍌 } = 3$$

Backward logic

$$\text{ 🍌 } = (4)^{-1} \times 12 = 3$$

Inverse definition

$$x^{-1} \times x = x \times x^{-1} := 1^1$$

Fundamental rule of algebra

You can apply any operation *to the same sides* of the equation

¹1 here is more generally the *multiplicative identity*

How do we do this for matrices?

Forward

$$4 \times 3 = 12$$

Backward

$$4 \text{ 🍌 } = 12 \Rightarrow \text{🍌} = 3$$

Backward logic

$$\text{🍌} = (4)^{-1} \times 12 = 3$$

Forward

$$A \times \underline{x} = \underline{y}$$

Backward¹

$$A\underline{x} = \underline{y}$$

$$\Rightarrow A^{-1} A\underline{x} = A^{-1} \underline{y}$$

$$\Rightarrow \underline{x} = A^{-1} \underline{y}$$

²

¹ If there is such a thing as the matrix inverse

² Only **left** multiplication legal here. $\underline{x} \neq \underline{y}A^{-1}$

Inverses sometimes exist for square matrices

$$A\underline{x} = \underline{y}$$

$$\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \underline{x} = \begin{bmatrix} 12 \\ 5 \end{bmatrix}$$

$$A^{-1}A\underline{x} = A^{-1}\underline{y}$$

$$\underline{x} = A^{-1}\underline{y}$$

```
np.linalg.inv(A):  
 $A^{-1} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$ 
```

Check

$$\begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} = \underline{I}_2$$

Verify: $\underline{x} = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 12 \\ 5 \end{bmatrix}$

Related question:

$$\begin{bmatrix} \text{£}3.00/\text{kg} & \text{£}2.00/\text{kg} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \text{ kg} \\ x_2 \text{ kg} \end{bmatrix} = \begin{bmatrix} \text{£}12 \\ 5\text{kg} \end{bmatrix}$$

$x_1 = \text{🍏}, x_2 = \text{🍌}$

Inverses don't always exist 😞

Numbers

- 0 has no inverse:

No solution

$$0 \times x = 20$$

Infinite solutions

$$0 \times x = 0$$

“Non-invertible” matrices behave like zero

Non-invertible (square) matrices

A square matrix is non-invertible if there exists a **nonzero** solution to $A\underline{x} = \underline{0}$

Example

$$\begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Consequence

When solving $A\underline{x} = \underline{y}$ to find $\underline{x}$, either

No solution

- e.g. $A\underline{x} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$.
- y_2 must be $\frac{y_1}{3}$. Why?

Infinite solutions

- ...but why?...

Back to the grocery problem

Questions

prices $\rightarrow \begin{bmatrix} \text{£}1.50 & \text{£}0.75 \end{bmatrix}$
weights $\rightarrow \begin{bmatrix} 0.5\text{lb} & 0.25\text{lb} \end{bmatrix}$

Answers

	Alice	Bob
cost?	£6	£4.50
weight?	2 lb	1.5 lb

What was in baskets?

Solution satisfies: $\begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 6 & 4.50 \\ 2 & 1.5 \end{bmatrix}$

Alternative solution:

$$\begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \left(\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} \right) = \begin{bmatrix} 6 & 4.50 \\ 2 & 1.5 \end{bmatrix}$$

Can you think of any other alternatives?

Back to the grocery problem

Alternative solution:

$$\begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \left(\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} + k \begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} \right) = \begin{bmatrix} 6 & 4.50 \\ 2 & 1.5 \end{bmatrix} \quad \forall k \in \mathbb{R}$$

- A solution for every $k \in \mathbb{R}$ so *infinite* solutions

General statement (square $A \in \mathbb{R}^{n \times n}$)

- Suppose exists nonzero vector $\underline{v} \in \mathbb{R}^n$ such that $A\underline{v} = \underline{0}$
- Then if $\underline{x} \in \mathbb{R}^n$ solves $A\underline{x} = \underline{y}$ for some fixed $\underline{y}$ so too does $\{\underline{x} + k\underline{v}\} \quad \forall k \in \mathbb{R}$

$$\begin{bmatrix} 1.50 & 0.75 \\ 0.5 & 0.25 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 6 & 4.50 \\ 2 & 1.5 \end{bmatrix} \quad \forall k \in \mathbb{R}$$

Why? The shop prices everything at £3/lb

- LHS: Row 2 = Row 1 / 3: *they ask the same question*
- RHS: Row 2 therefore *must* be row 1 / row 3.
- Bottom row is *useless*

$$\begin{bmatrix} 1.50 & 0.75 \\ \bullet & \bullet \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} 6 & 4.50 \\ \bullet & \bullet \end{bmatrix} \quad \forall k \in \mathbb{R}$$

- Two effective equalities (i.e. constraints)
- **Four** free variables (degrees of freedom)

Of course there are many solutions.

Tall or wide systems

$M \in \mathbb{R}^{m \times n}$: m questions, n unknowns

Square: $m = n$

$$\begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} \begin{bmatrix} \bullet \\ \bullet \end{bmatrix} = \begin{bmatrix} \bullet \\ \bullet \end{bmatrix}$$

Questions = unknowns

- **Independent** questions imply exactly one basket

Tall: $m > n$

$$\begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} \begin{bmatrix} \bullet \\ \bullet \end{bmatrix} = \begin{bmatrix} \bullet \\ \bullet \\ \bullet \end{bmatrix}$$

Questions > unknowns

- **Overconstrained** with independent questions: no solution

Wide: $m < n$

$$\begin{bmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \end{bmatrix} \begin{bmatrix} \bullet \\ \bullet \\ \bullet \end{bmatrix} = \begin{bmatrix} \bullet \\ \bullet \end{bmatrix}$$

Questions < unknowns

- Usually *infinite* possible answers
- Pick the **best** one: optimisation!

Next time: Matrices as transformations of
space¹

¹Humans are better at geometry (pictures) than symbols (numbers)

Next lecture

Practice: unit spheres norm $\rightarrow$ metric

correlation is also cosine Abstract definition of vector space and field Bilinearity of dot product?

Basis of a vector space Linear in/dependence of vectors Kernel/image What is a linear map?

Matrices represent linear maps Duality of square matrices: transform vs change of basis Matrix multiplication composes linear maps What is a rotation/scaling (practice) SVD?